

Practical No. 1

Aim: Study and installation of MySQL/Oracle

Software Requirement: MySQL, Oracle 10g.

Theory:

MySQL is an open-source, relational database management system (RDBMS) that uses Structured Query Language (SQL) to manage and manipulate data. It is one of the most popular database systems used in web applications, known for its speed, reliability, and ease of use. MySQL is commonly used in conjunction with programming languages such as PHP, Java, and Python to build dynamic websites and applications.

Key Features in MySQL

MySQL is a popular choice for managing **relational databases** for several reasons:

- **Open-Source:** MySQL is free and open-source, allowing modification and distribution.
- **High Performance:** It offers fast data retrieval and processing for large datasets.
- **Scalability:** Supports large databases and high traffic with features like partitioning and clustering.

Applications of MySQL

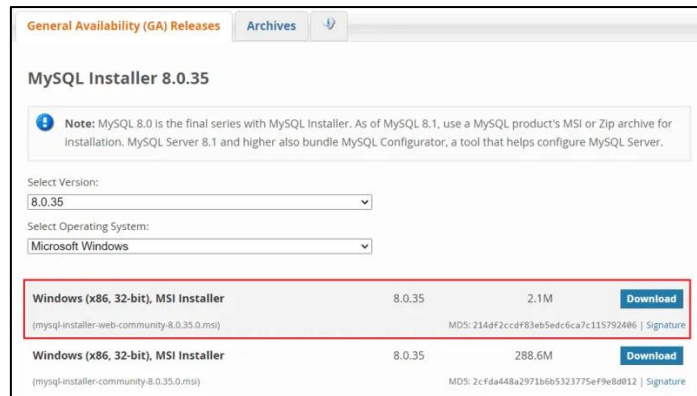
- **E-commerce:** MySQL is extensively used in e-commerce platforms for managing **product catalogs, customer data**, orders, and transactions.
- **Healthcare:** MySQL is used in **healthcare applications** for storing and managing **patient records**, medical histories, treatment plans, and diagnostic information.

Installation of MySQL:

Installing MySQL on Windows is a straightforward process that will allow you to utilize the power of one of the most popular **relational database management systems** (RDBMS).

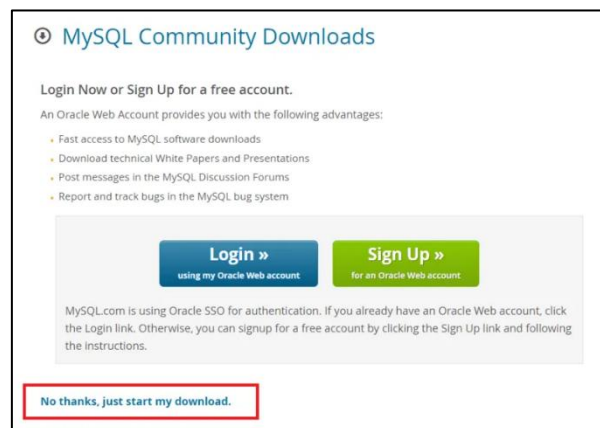
Step 1: Visit the Official MySQL Website

Open your preferred web browser and navigate to the official MySQL website. Now, Simple click on first download button.



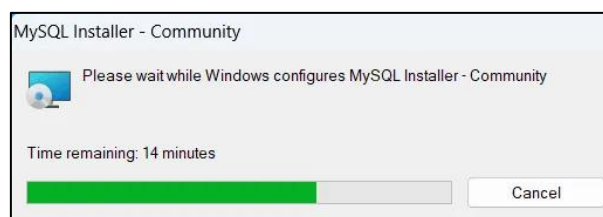
Step 2: Go to the Downloads Section

On the MySQL homepage, Click on the " **No thanks, just start my download**" link to proceed MySQL downloading.



Step 3: Run the Installer

After MySQL downloading **MySQL.exe** file , go to your Downloads folder, find the file, and double-click to run the installer.

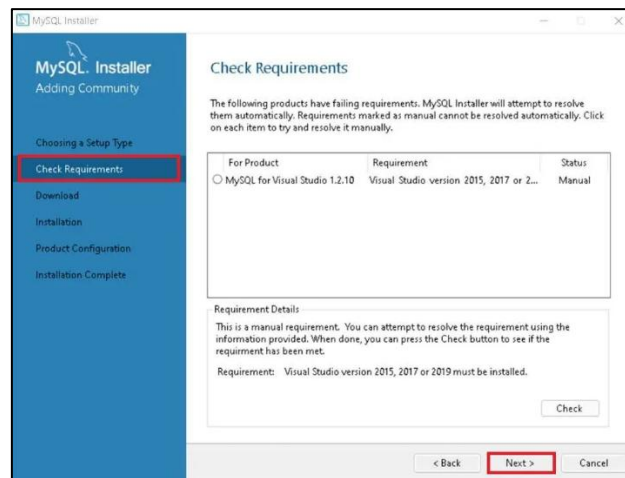


Step 4: Choose Setup Type

The installer will instruct you to choose the setup type. For most users, the "**Developer Default**" is suitable. Click "Next" to proceed.

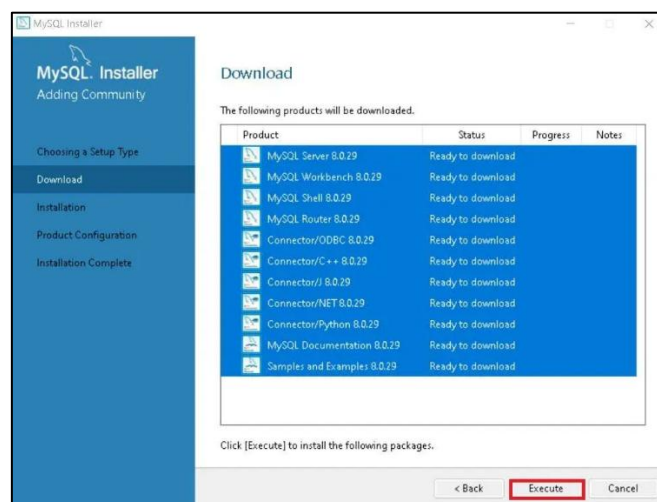
Step 5: Check Requirements

You might be prompted to install necessary MySQL software, typically **Visual Code**. The installer can auto-resolve some issues, but not in this case.



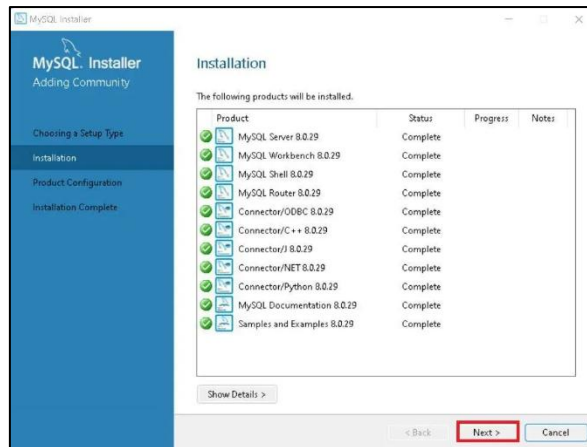
Step 6: MySQL Downloading

Now that you're in the download section, click "**Execute**" to start downloading the components you selected. Wait a few minutes until all items show tick marks, indicating completion, before moving forward.



Step 7: MySQL Installation

Now the downloaded components will be installed. Click "**Execute**" to start the installation process. MySQL will be installed on your Windows system. Then click **Next** to proceed

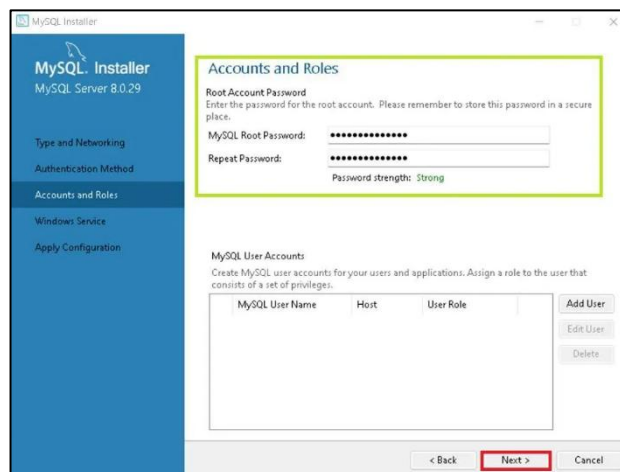


Step 8: Navigate to Few Configuration Pages

Proceed to **"Product Configuration" > "Type and Networking" > "Authentication Method"** Pages by clicking the "Next" button.

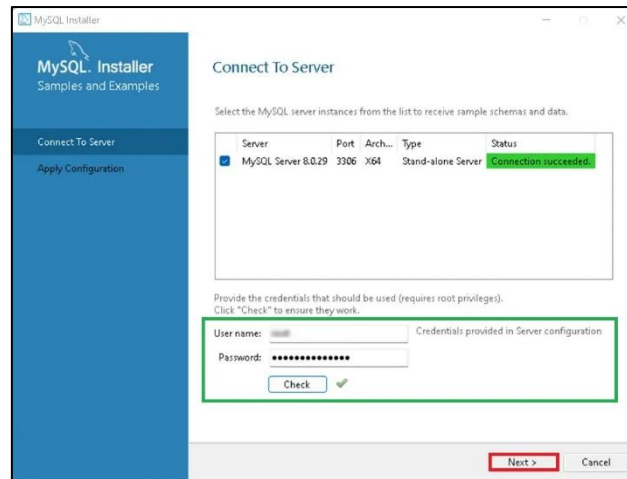
Step 9: Create MySQL Accounts

Create a password for the MySQL root user. Ensure it's strong and memorable. Click **"Next"** to proceed.



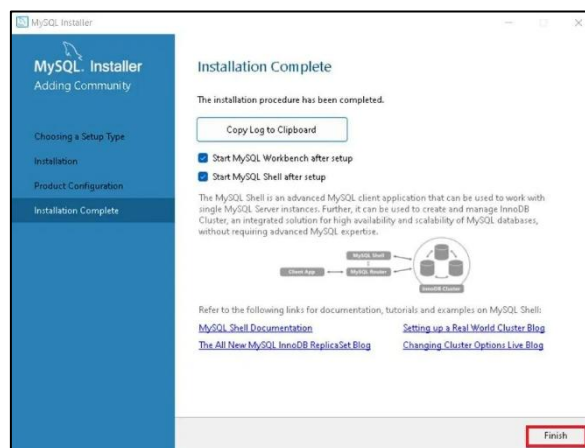
Step 10: Connect To Server

Enter the root password, click Check. If it says **"Connection succeed,"** you've successfully connected to the server.



Step 11: Complete Installation

Once the installation is complete, click "**Finish.**" Congratulations! MySQL is now installed on your Windows system.



Post-Installation Steps

Once MySQL is successfully installed, you can manage and interact with your databases in several ways:

- **Create Databases and Tables:** You can now create new databases using the MySQL Workbench interface or via the command line using SQL commands.

```
MySQL 8.0 Command Li x + v
Enter password: ****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 24
Server version: 8.0.42 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database admin;
Query OK, 1 row affected (0.01 sec)

mysql> use admin;
Database changed
mysql> create table books (id int, book_name char(50), authors char(50), publishers char(50));
Query OK, 0 rows affected (0.02 sec)

mysql> insert into books values(1, 'Study in Scarlet', 'Sherlock Holmes', 'pubslier');
Query OK, 1 row affected (0.00 sec)

mysql> insert into books values(2, 'The Sign of the Four', 'Sherlock Holmes', 'pubslier');
Query OK, 1 row affected (0.00 sec)

mysql> insert into books values(3, 'Valley of Death', 'Sherlock Holmes', 'pubslier');
Query OK, 1 row affected (0.01 sec)

mysql> insert into books values(4, 'Hound of Baskervilles', 'Sherlock Holmes', 'pubslier');
Query OK, 1 row affected (0.01 sec)

mysql> select * from books;
+----+-----+-----+-----+
| id | book_name | authors | publishers |
+----+-----+-----+-----+
| 1 | Study in Scarlet | Sherlock Holmes | pubslier |
| 2 | The Sign of the Four | Sherlock Holmes | pubslier |
| 3 | Valley of Death | Sherlock Holmes | pubslier |
| 4 | Hound of Baskervilles | Sherlock Holmes | pubslier |
+----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

Conclusion: Hence, from the above steps we have successfully studied and installed MySQL.